

CALCULUS 101 · WEEK 1 QUIZ

Calculus 101 — Week 1

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. Calculus is the mathematics of...

- ☐ A. shapes and angles, like triangles, circles, and cubes.
- ☐ B. chance — how likely something is to happen.
- ☐ C. counting very large numbers, like millions and trillions.
- ☐ D. how things change — how fast, and how much piles up over time.

2. A baby weighed 900 g on day 1 and 970 g on day 8. What was the rate of change?

- ☐ A. 7 grams per day
- ☐ B. 70 grams per day
- ☐ C. 100 grams per day
- ☐ D. 10 grams per day

3. On a graph of weight over time, what does a STEEPER line mean?

- ☐ A. The baby weighs more than the other babies.
- ☐ B. The baby is older than the other babies.
- ☐ C. The weight is changing faster.
- ☐ D. The days are passing more quickly.

4. A flat, level line on a weight graph means...

- ☐ A. the baby is losing weight slowly.
- ☐ B. the baby is gaining weight slowly.
- ☐ C. the weight isn't changing at all.
- ☐ D. the scale is not working properly.

5. Most newborns LOSE a little weight in their first few days. What does that look like on the graph?

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- A. The line goes downhill – a negative rate of change.
- B. The line stays flat, because losing weight doesn't count.
- C. The line goes uphill, just more slowly than usual.
- D. The line jumps straight up when the weight comes back.

6. Who invented calculus?

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- A. Antonie van Leeuwenhoek, to keep count of the cells he saw in his microscope.
- B. Virginia Apgar, as part of creating her scoring system for newborn babies.
- C. Isaac Newton and Gottfried Leibniz, working separately at about the same time.
- D. Joseph Priestley, to measure the gases coming off his chemistry experiments.

7. A doctor writes that a baby is gaining “20 g/kg/day.” What does that mean?

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- A. The baby gains 20 grams once every 20 days, whatever its size.
- B. Each kilogram of the baby gains about 20 grams every day.
- C. The whole baby gains a total of 20 kilograms every day.
- D. The baby currently weighs 20 kilograms and is gaining daily.

8. Why do NICU doctors judge growth by the trend line over several days rather than one day's weight?

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- A. A baby has to be weighed at least ten times before any of the weights is allowed to count.
- B. One day's weight can jump from fluid or a diaper; the line over days shows real growth.
- C. The scale only gives a trustworthy reading after it has been used several days in a row.
- D. Single weights are always rounded to the nearest ten grams, so none of them is accurate.

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)